PROYECT

“FAKE NEWS DETECTOR”

Work Report

Bologna, February 2026

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1 Introduction

1.1 Background to the problem

In recent years, the proliferation of information on the internet and social media has facilitated the rapid spread of news on a global scale. However, this phenomenon has also led to a significant increase in so-called ‘fake news’, which can generate misinformation, influence public opinion and affect critical areas such as politics, the economy and public health.

The detection of fake news has become a major issue within the field of natural language processing (NLP), as it involves analysing large volumes of text and automatically determining its veracity. Unlike other classification problems, the detection of fake news presents an added complexity, as fake news can mimic the style and structure of real news, making it difficult to identify using simple techniques.

In this context, the development of automated systems capable of distinguishing between real and fake news represents both a technical and a social challenge, and constitutes an active area of research in artificial intelligence.

1.2 Project Objective

The main objective of this project is to develop a system capable of automatically classifying news as true or false using natural language processing techniques and deep learning models.
To achieve this overall objective, the following specific objectives have been set:

- To implement a baseline text classification model using classical techniques such as TF-IDF and logistic regression, in order to establish an initial benchmark.
- To develop more advanced models based on transformer architectures, in particular the BERT model, using fine-tuning techniques.
- To analyse the impact of different text representations, comparing the exclusive use of the headline with the combined use of the headline and the full content of the news item.
- To experiment with different fake news datasets, such as WELFake and ISOT, evaluating their characteristics and potential biases.
- Detects and corrects data leakage issues and other biases present in the data, with the aim of improving the model's generalisation ability.
- Evaluate the performance of the models using standard metrics such as accuracy, precision, recall and F1-score.

Taken together, these objectives enable us not only to build a functional system, but also to gain an in-depth understanding of the challenges associated with the automatic detection of fake news.

1.3 Technology used

COMPONENT	LIBRARY
Language	Python
NLP	Transformers (HuggingFace)
Deep Learning	PyTorch
Data analysis	Numpy, Pandas, Matplotlib, Seaborn, Scipy, Wordcloud
Machine Learning	Scikit-learn
Model	BERT
Datasets	Datasets (HuggingFace)

2 State Of The Art

2.1 Fake news detection in NLP

The automatic detection of fake news is a significant challenge within the field of natural language processing (NLP). This challenge involves analysing journalistic texts and determining whether their content is accurate or has been manipulated for the purpose of spreading misinformation.

Unlike other text classification tasks, the detection of fake news is more complex, as fake news often mimics the linguistic style, structure and tone of real news. This makes it difficult to identify using simple techniques based solely on keywords or superficial patterns.

Traditionally, approaches to tackling this problem have fallen into two broad categories: classical methods based on linguistic features and more advanced models based on deep learning. As the field of NLP has evolved, methods have progressively incorporated techniques capable of better capturing the meaning and context of language.

2.2 Models used in the literature

Firstly, there are the traditional models, which convert text into numbers using techniques such as TF-IDF. These values are then used in simple models such as logistic regression to determine whether a news story is true or false. These methods are quick and easy to use, but they have a major drawback: they do not understand the meaning of the text; they merely count words.

Later on, models based on neural networks, such as RNNs or LSTMs, began to be used. These models analyse the text as a sequence of words, which allows them to grasp the context better than traditional methods. However, they are slower and struggle when dealing with long texts.

Currently, the most advanced models are those based on transformers, such as BERT. These models represent the state of the art in NLP because they can better understand the full context of a sentence. Instead of analysing words one by one, they process the entire sentence at once, enabling them to better grasp the meaning of the text.

For this reason, in this project we have opted to use transformer-based models, as they offer better results in text classification tasks such as fake news detection.

2.3 Introduction to BERT

BERT (Bidirectional Encoder Representations from Transformers) is a language model developed by Google based on the Transformer architecture. Its key feature is that it analyses the context of words in a bidirectional manner, i.e. by taking into account both preceding and following words within a sentence. This represents a significant improvement over previous models, which processed text sequentially.

BERT has been pre-trained on large volumes of text, enabling it to acquire a general understanding of language. It can subsequently be adapted to specific tasks, such as text classification, through a process known as fine-tuning. This process involves adjusting the model's parameters using a specific dataset, in this case, real and fake news stories.

In this project, the base version of the model, known as bert-base-uncased, has been used; this works with English text without distinguishing between upper and lower case. This model has been trained and evaluated on various fake news datasets, allowing us to analyse its performance and its ability to generalise across different scenarios.

The choice of BERT is justified by its high performance in NLP tasks and its availability through libraries such as HuggingFace, which facilitates its implementation and experimentation.

3 Datasets Used

Various news datasets have been used in the development of this project, with the aim of training and evaluating models capable of distinguishing between real and fake news. Specifically, the WELFake and ISOT datasets have been used, both of which are widely employed in fake news detection tasks.

The use of multiple datasets allows us to analyse the model's performance in different contexts and improve its ability to generalise.

3.1 Dataset WELFake

The WELFake dataset contains a large collection of news articles labelled as real or fake. Each entry consists mainly of:

- title: the title of the news article
- text: the full content
- label: the label (0 = real, 1 = fake)

This dataset is useful because it combines different sources of information, making it more varied. However, it may also contain certain patterns that the model might learn superficially rather than truly understanding the content.

3.2 Dataset ISOT

The ISOT dataset consists of two main files:

- True.csv → real news articles
- Fake.csv → fake news articles

Each news article includes:

- Title
- Full Text

This dataset is widely used in research, but it has a significant problem: many real news articles include references to the “Reuters” news agency, which introduces a bias into the data.

3.3 Problems detected in the datasets

During the analysis of the datasets, biases were detected that affect the model’s performance.
In the case of the ISOT dataset, many real news articles contain expressions such as:

“(Reuters) -”

The model may learn to identify this word as a sign that the news item is real, without analysing the content. This phenomenon is known as data leakage, as the model uses irrelevant information to make decisions.

4 Data Preprocessing

4.1 Data cleansing

Before training the models, the data must be prepared to avoid errors and improve its quality. Firstly, rows containing null values in text fields, such as the headline or the body of the news article, were removed.

This step is important, as language models cannot process incomplete data and may generate errors during training.

4.2 Bias elimination (ISOT)

During the analysis of the ISOT dataset, it was found that many real news articles included references to the “Reuters” agency, such as:

“WASHINGTON (Reuters) -”

This type of information can introduce bias, as the model might learn to identify these words as a sign that the news article is real, without analysing its content.

To avoid this problem, a cleaning process was applied using regular expressions, removing these references and replacing them with more neutral terms. In this way, the model is forced to base its decisions on the content of the news item rather than on superficial patterns.

5 Methodology

5.1 Splitting the dataset

To train and evaluate the models, the dataset was split into two parts:

- 80% for training
- 20% for testing

In addition, a stratified split was used, ensuring that both the training and test sets contain a similar proportion of real and fake news. This allows for a more reliable evaluation of the model.

5.2 Models implemented

Two types of models have been used in this project for news classification:

- ### 5.2.2 Baseline

  As a starting point, a simple model was implemented based on:

  - TF-IDF to convert the text into numerical values
  - Logistic regression to perform the classification

  This model serves as an initial benchmark for comparing the performance of more advanced models. Although it is quick and easy to implement, it has limitations because it does not understand the context of the language.

- ### 5.2.3 BERT model

  The main model used in the project is BERT (bert-base-uncased), based on the Transformer architecture.

  The process followed was as follows:

  1. **Text tokenization**
     The text is converted into a format that the model can understand using a tokeniser.
  2. **Conversion to tensors**
     The data is converted into numerical structures compatible with PyTorch.
  3. **Training (fine-tuning)**
     The pre-trained model is fine-tuned using the project's dataset over several epochs.
  4. **Prediction**
     Once trained, the model is used to classify new news stories as true or false.

6 System Implementation

6.1 Project structure

To organise the development in a clear and modular way, the project has been divided into several directories within the src directory, each with a specific purpose:

- data → data loading and pre-processing
- models → model training and deployment
- evaluation → metric calculation and results analysis
- utils → utility functions

This structure allows for the separation of responsibilities and facilitates code reusability.

6.2 Data module (src/data_loader)

This module is responsible for preparing the datasets prior to training. It includes:

- loading files (CSV or external datasets)
- data cleaning (removal of missing values)
- text pre-processing

- combining title and content

Its aim is to generate data ready for use by the models.

6.3 Models module (src/modelling)

This module implements the logic related to classification models.
It includes:

- definition of the BERT model
- text tokenization
- creation of the dataset for PyTorch
- training using the transformers library
- generation of predictions

This module allows models to be trained and reused in an organised manner.

6.4 Evaluation module (src/evaluation)

This module is responsible for analysing the performance of the models.
It includes:

- calculation of metrics (accuracy, precision, recall, F1-score)
- generation of results reports
- confusion matrices
- comparison between different models

It allows for a clear evaluation of the system's quality.

6.5 Utilities module (src/eda_utils)

This module includes auxiliary functions designed primarily for data analysis and visualisation.
It enables tasks such as:

- visualising the distribution of real and fake news
- generating word clouds
- analysing specific terms
- comparing text lengths
- representing experimental results

Its aim is to facilitate the interpretation of the data and the results obtained, whilst avoiding duplication of code in other parts of the project.

7 Experiments Carried Out

7.1 Ablation study

The ablation study was conducted with the aim of analysing the impact of the information used by the model on the classification task.
Specifically, two configurations were compared:

- Use of title only
- Use of title + full content

This analysis allows us to verify whether the model performs better when given more context. The results show that using the full text provides better performance, as it allows the model to better understand the meaning of the news item.
Example of BERT using the WELFake dataset:

```
Training BERT Expert: TITLES_ONLY (Max Length: 128)
Loading weights: 100% 199/199 [00:00<00:00, 3182.25it/s]
```

Epoch	Training Loss	Validation Loss	Accuracy
1	0.285272	0.248587	0.909000
2	0.145639	0.267352	0.927000

```
Training BERT Expert: FULL_CONTENT (Max Length: 256)
Loading weights: 100% 199/199 [00:00<00:00, 5366.08it/s]
```

Epoch	Training Loss	Validation Loss	Accuracy
1	0.159376	0.089730	0.979500
2	0.028946	0.070486	0.985500

7.2 Model comparison

This project compared two types of models:

- Baseline model (TF-IDF + Logistic Regression)
- BERT model

Although the baseline model is quick and simple, it has limitations as it does not understand the context of the language. On the other hand, BERT is capable of capturing more complex semantic relationships, which translates into better performance.

The results show that BERT outperforms the baseline model on all metrics, particularly when using the full text of the news articles.

7.3 Experiments using different datasets

To assess the model's generalisation ability, experiments were conducted using different datasets:

- WELFake
- ISOT

The use of multiple datasets made it possible to analyse how the model's performance varies depending on the data used.

After applying cleaning techniques to remove these biases, a more realistic performance was achieved, confirming that the model learns the content of the news more effectively rather than simply memorising patterns.
Test in a real news story:

index	Expert	Mode	Confidence	Label
0	WELFake (Titles)	Title Only	0.9057996869087219	FAKE
1	WELFake (Full)	Full Context	0.999038815498352	TRUE
2	ISOT (Titles)	Title Only	0.94503253698349	TRUE
3	ISOT (Full)	Full Context	0.9996706247329712	TRUE
4	Fusion (Titles)	Title Only	0.9047427177429199	FAKE
5	Fusion (Full)	Full Context	0.9998835325241089	TRUE

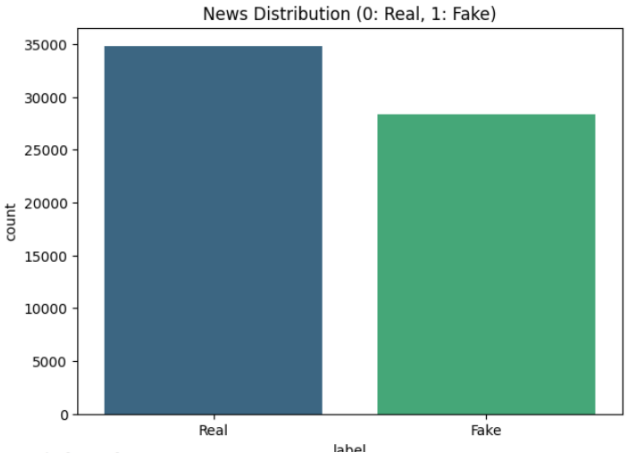
8 Results

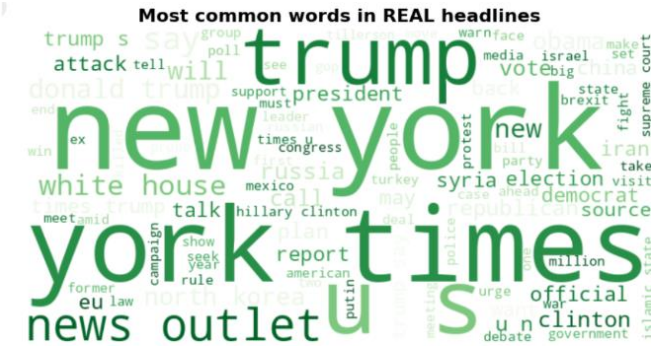
8.1 Metrics obtained

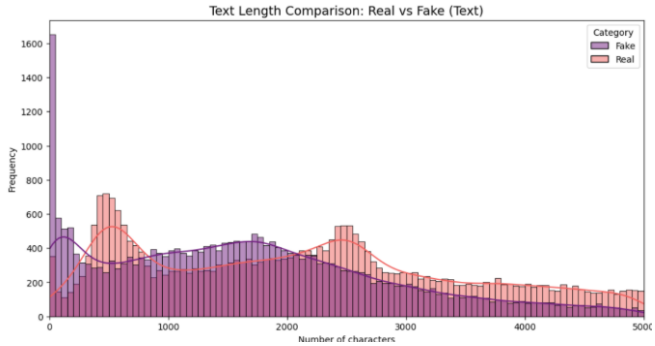
The results were analysed in two phases:

- Exploratory data analysis (EDA):
  The distribution of real and fake news stories and the most frequent words were examined using graphs and word clouds. This provided a better understanding of the data and helped to identify potential patterns.
- Training:
  Accuracy was the primary metric used.
  The results show that:
  - BERT achieves better results than the baseline
  - Using full text yields better results than using headlines alone

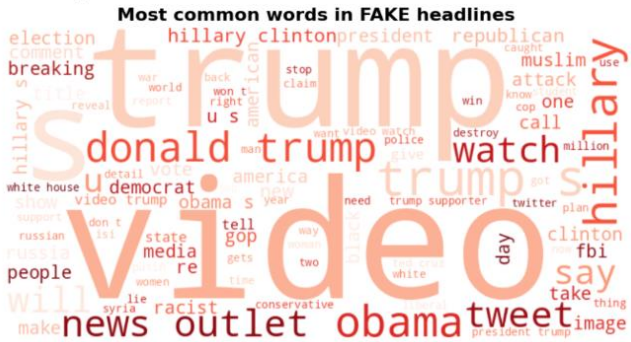
Data Analysis WELFake







--- Generating Word Cloud for Fake news ---



8.2 Comparison of models

The following were compared:

- Baseline
- BERT (headings)
- BERT (full text)

Results:

- BERT outperforms the baseline
- Full text improves performance
- Context is key

However, high accuracy does not always translate to good performance in real-world scenarios.

Baseline model, True new

index	Expert	Mode	Confidence	Label
0	WELFake Specialist	Title Only	0.9581950895976535	FAKE
1	WELFake Specialist	Body Text Only	0.5751839391475652	TRUE
2	WELFake Specialist	Full Context	0.6331106296095692	FAKE
3	ISOT Specialist	Title Only	0.5919973811538404	TRUE
4	ISOT Specialist	Body Text Only	0.568386527041916	TRUE
5	ISOT Specialist	Full Context	0.6928032613100354	TRUE
6	Hybrid Specialist	Title Only	0.8606591240690583	FAKE
7	Hybrid Specialist	Body Text Only	0.6037362894868321	TRUE
8	Hybrid Specialist	Full Context	0.5280975688683636	TRUE

BERT model, True new

index	Expert	Mode	Confidence	Label
0	WELFake (Titles)	Title Only	0.9057996869087219	FAKE
1	WELFake (Full)	Full Context	0.999038815498352	TRUE
2	ISOT (Titles)	Title Only	0.94503253698349	TRUE
3	ISOT (Full)	Full Context	0.9996706247329712	TRUE
4	Fusion (Titles)	Title Only	0.9047427177429199	FAKE
5	Fusion (Full)	Full Context	0.9998835325241089	TRUE

9 Analysis Of Results

The baseline model performs the worst as it does not take the text context into account.
The BERT model improves the results, and its version using the title plus the full text is the most accurate, as it utilises more information.
Overall, the results show that using more advanced models and a greater amount of context clearly improves classification.

9.1 The problem of overfitting

During the experiments, very high accuracy scores (close to 98–99%) were observed, which might initially suggest that the model performed exceptionally well.

However, when testing the model on external real-world news, it was found that in some cases it failed to classify correctly. This suggests the presence of overfitting; in other words, the model has learnt the training data too well, but does not generalise correctly to new data.

This problem is due, in part, to the limited size of the dataset used and the presence of repetitive patterns.

9.2 Data leakage

One of the main issues identified was data leakage in the ISOT dataset.

It was observed that many real news articles included the word “Reuters”, which the model learned to treat as a direct indicator of veracity. Instead of analysing the content, the model used such patterns as shortcuts for classification.

To mitigate this problem, a pre-processing stage was applied to remove these references using regular expressions. Following this cleaning, the model's performance decreased slightly, but it became more realistic and reliable.

9.3 Generalisation of the model

Although the model performs well on the test data, its performance declines when applied to real-world news outside the dataset.

This suggests that its generalisation ability is limited, due to:

- differences between datasets
- biases in the training data
- learning superficial patterns

10 Limitations of the project

The project has certain limitations that affect the model's performance and generalisation ability.

Firstly, the size of the dataset and the use of a limited number of examples may contribute to overfitting issues, reducing the model's ability to perform correctly with external real-world news.

Furthermore, the model used (BERT base) has limited capabilities compared to more advanced models, which may affect its performance on complex tasks.

There are also limitations arising from the use of Google Colab, such as variable GPU availability, memory constraints and limited execution time, which necessitates working with subsets of data and reduced configurations.

Finally, the system has been evaluated primarily using data similar to the training data, so its behaviour in real-world scenarios may vary.

11 Future Improvements

There are several possible improvements that could enhance the performance and robustness of the system.

First, it would be beneficial to use larger and more diverse datasets, combining multiple sources to reduce bias and improve the model's generalization ability.

It would also be interesting to explore more advanced models, such as improved versions of BERT or more recent architectures, which offer better language understanding capabilities.

Another important improvement would be to optimize the training process by tuning hyperparameters (such as epochs, batch size, and sequence length) and applying more advanced *data augmentation* techniques.

Additionally, the system could be extended to analyze real-time news, integrating the model into an application or user interface.

Finally, using more powerful computational resources would allow training with larger datasets and more complex configurations, leading to better overall performance.

12 Conclusions

In this project, a system for fake news detection was developed and evaluated using both traditional and deep learning approaches.

The results show that transformer-based models, such as BERT, significantly outperform baseline methods, especially when using the full text of the news. This confirms the importance of contextual information in natural language processing tasks.

However, the analysis also revealed important limitations, such as dataset bias, overfitting, and limited generalization to real-world data. These findings highlight that high accuracy alone is not sufficient to guarantee reliable performance.

Overall, the project demonstrates the potential of machine learning techniques for fake news detection, while also emphasizing the need for careful data selection and evaluation.

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- PyTorch Documentation: https://pytorch.org/